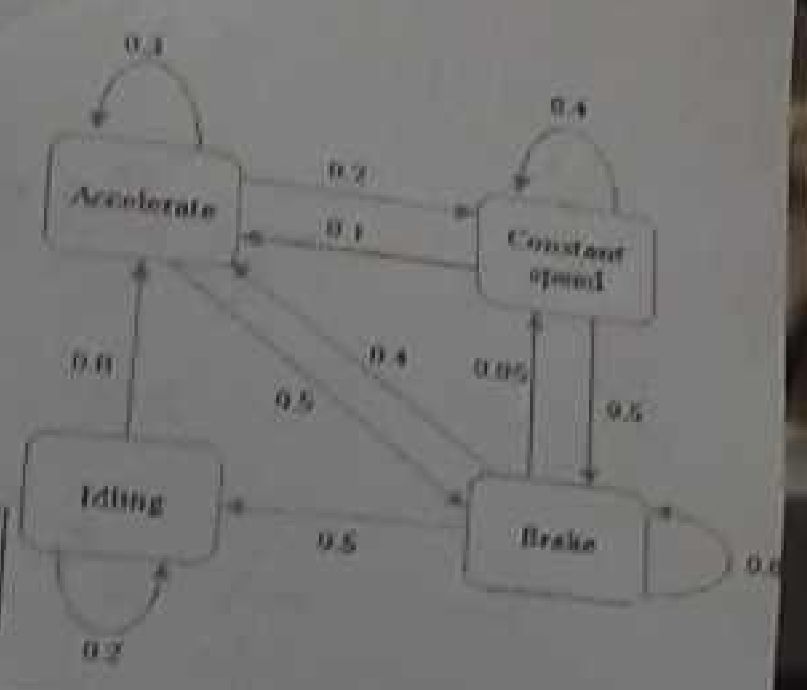
Course Code: 3201; Title: Mathematical Analysis for Computer Science.

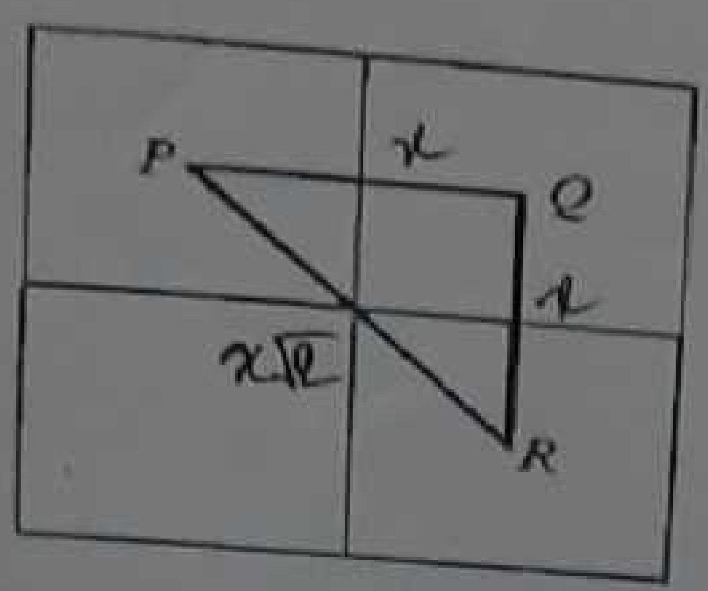
1. Define recursion. Write down the master theorem of recursion for GCD and Tower of Hanoi. (2)
2. Write down the differences between divergent and convergent. Is harmonic series divergent or convergent? Explain. (2)
3. Define marcov chain along with its uses. Find the transition matrix for the following graph.(2)



Ques: 03

Marks for Ques 5. - (4)

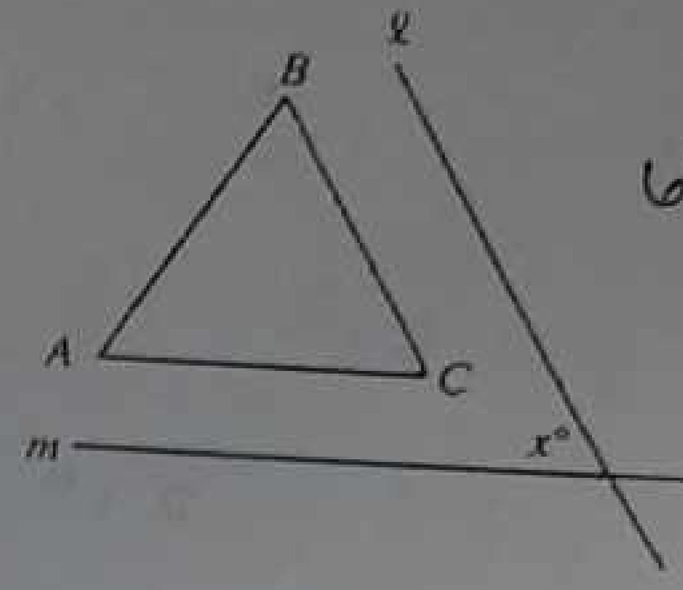
4. (a)- (1)



$2x+x\sqrt{2}$

In the figure above, each of the four squares has sides of length x. If $\triangle PQR$ is formed by joining the centers of three of the squares, what is the perimeter of $\triangle PQR$ in terms of x?

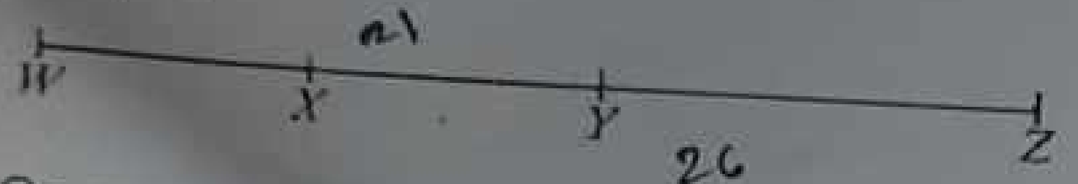
4. (b) What is the value of x? –(1)



60°

$\triangle ABC$ is equilateral. Line ℓ is parallel to side BC and line m is parallel to side AC.

4. (c) - (0.5)



On segment WZ above, if $WY = 21$, $XZ = 26$, and YZ is twice WX, what is the value of XY?

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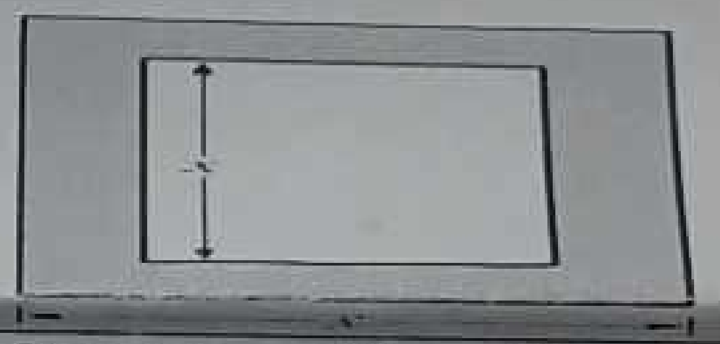
4. (d) – (1)

A flat rectangular picture, represented by the unshaded region in the figure above, is mounted in a flat rectangular frame, represented by the shaded region. The frame is 1 inch wide on all sides. For what value of x, in inches, is the area of the frame equal to the area of the picture?

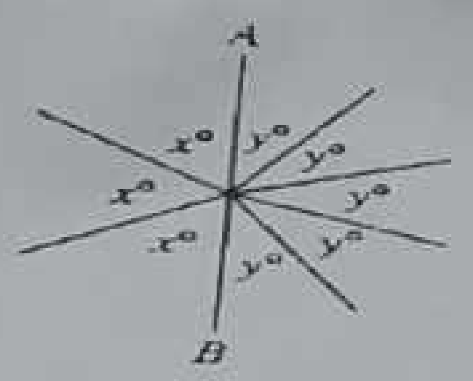


6

$w \times 2\pi r = \pi r^2$

$\Rightarrow w = r/2$

4.(e) (0.5)



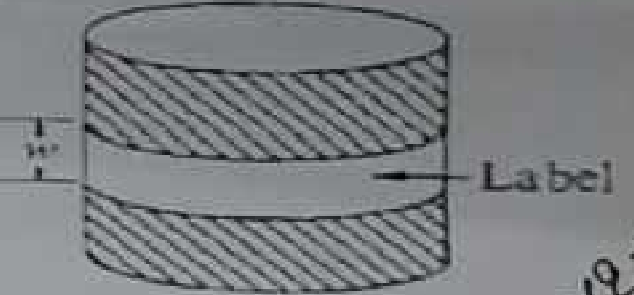
$x = 60°$, $y = 36$

$\frac{1}{9}$

In the figure above, AB is a line segment. What is the value of $\frac{x-y}{x+y}$?

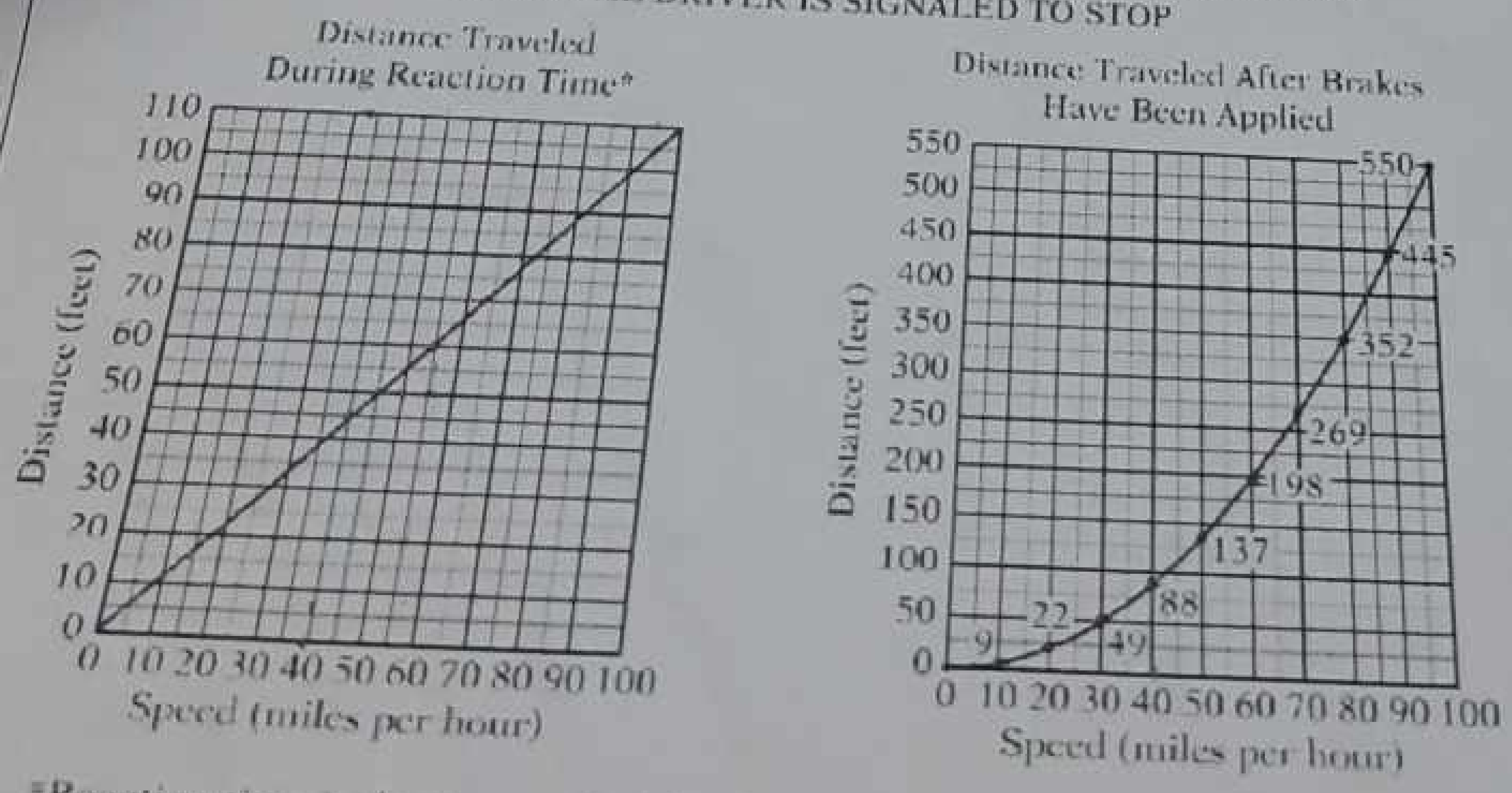
4. (f) Compare w eith r. –(1)

$w = r/2$



A rectangular label is attached to a right circular cylinder with radius r. The label, which encircles the cylinder without overlap, has width w and an area equal to the area of the base of the cylinder.

DISTANCE TRAVELED BY A CAR ACCORDING TO THE CAR'S SPEED WHEN THE DRIVER IS SIGNALED TO STOP



*Reaction time is the time period that begins when the driver is signaled to stop and ends when the driver applies the brakes.

Note: Total stopping distance is the sum of the distance traveled during reaction time and the distance traveled after brakes have been applied.

5.(a) The speed, in miles per hour, at which the car travels a distance of 52 feet during reaction time is closest to which of the following?
i. 43 [✓ ii. 47] iii. 51 iv. 55 v. 59

(b) Approximately what is the total stopping distance, in feet, if the car is traveling at a speed of 40 miles per hour when the driver is signaled to stop?
[✓ i. 130] ii. 110 iii. 90 iv. 70 v. 40

(c) Of the following, which is the greatest speed, in miles per hour, at which the car cantravel and stop with a total stopping distance of less than 200 feet?
[✓ i. 50] ii. 55 iii. 60 iv. 65 v. 70

5. (d)The total stopping distance for the car traveling at 60 miles per hour is approximately what percent greater than the total stopping distance for the car traveling at 50 miles per hour? i. 22% ii. 30% [✓ iii. 38%] iv 45% v. 52%

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